DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING • B.TECH DEGREE PROGRAM

FUNDAMENTALS OF DATA SCIENCE (FDS)

Academic Case Study Report • Batch 3 (Case Study 3)

Predicting Titanic Passenger Survival Using Logistic Regression: Exploratory Analysis, Leakage-Free Pipeline Design, Statistical Odds-Ratio Interpretation, and Streamlit Deployment

An Empirical Binary Classification Study on the Faculty-Supplied RMS Titanic Manifest Dataset (N = 891)

Candidate & Course Submission Details (Prescribed Format)

Prescribed Field	Submission Details / Placeholder for Student Completion
Title of Case Study	Case Study 3: Predicting Titanic Passenger Survival Using Logistic Regression
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Reproducibility & Academic Integrity Declaration: All statistical metrics, visualizations, confusion matrices, ROC curves, and parameter estimates in this report were computed directly by executing the accompanying Python scripts (`src/train_model.py`, `app.py`) and Jupyter Notebook (`notebooks/titanic_logistic_regression.ipynb`) on the faculty-supplied dataset.

2. Abstract

This case study investigates the application of binary Logistic Regression to predict passenger survival aboard the RMS *Titanic* using the faculty-supplied dataset of 891 passenger records and 12 variables. First, an independent cross-format audit of the provided CSV (`titanic.csv`) and Excel (`titanic (1).xlsx`) files confirmed identical records and missingness patterns (`Cabin`: 77.10% missing, `Age`: 19.87% missing, `Embarked`: 0.22% missing), while identifying a mixed integer-string data type discrepancy in the Excel `Ticket` column. Second, exploratory data analysis revealed pronounced bivariate survival disparities across biological sex (74.20% female survival vs. 18.89% male survival) and socioeconomic ticket class (62.96% in 1st class, 47.28% in 2nd class, and 24.24% in 3rd class). Third, a leakage-free `scikit-learn` `Pipeline` and `ColumnTransformer` architecture was constructed using stratified 80/20 train-test splitting (712 training records, 179 test records, `random_state=42`), training-only median/mode imputation, standardization, and one-hot encoding, achieving 5-fold cross-validation Accuracy of **79.49% (±2.73%)**, ROC-AUC of **0.8549 (±0.0212)**, and F1-score of **0.7246 (±0.0335)** on the training set. On the held-out test set, the L2-regularized logistic regression classifier achieved an Accuracy of **80.45%** (144/179), Precision of **79.31%**, Recall of **66.67%**, Specificity of **89.09%**, F1-score of **0.7244**, and an ROC-AUC score of **0.8433**. Multivariable odds-ratio and Wald z-test analyses confirmed that male sex ($\beta = -2.5600$, OR = 0.0773, $p < 0.0001$), lower ticket class ($\beta = -0.9158$, OR = 0.4002, $p < 0.0001$), older age ($\beta = -0.5007$, OR = 0.6061, $p < 0.0001$), and larger sibling/spouse counts ($\beta = -0.2573$, OR = 0.7731, $p = 0.0396$) significantly reduce survival odds when holding other variables constant. Finally, the trained pipeline was integrated into an interactive Streamlit web application (`app.py`) for real-time passenger survival prediction.

3. Introduction

The sinking of the RMS *Titanic* on April 15, 1912, following a collision with an iceberg in the North Atlantic, remains one of the most extensively studied maritime disasters in history. Because lifeboat capacity was sufficient for only approximately half of the individuals on board, survival outcomes were strongly shaped by institutional evacuation norms—most notably the maritime principle of *"women and children first"*—as well as structural barriers linked to cabin deck depth and ticket class. In data science education and statistical modeling, the Titanic passenger manifest serves as a foundational benchmark for supervised binary classification because it combines continuous numerical variables, ordinal socioeconomic tiers, nominal categorical attributes, skewed monetary distributions, and realistic missing data.

Unlike black-box ensemble or deep learning algorithms, **Logistic Regression** provides a mathematically transparent probabilistic framework grounded in generalized linear modeling. By linking a linear combination of predictors to the Bernoulli probability of survival through the logistic (sigmoid) function, logistic regression yields both calibrated class probabilities and directly interpretable multiplicative odds ratios. This report documents the complete data science lifecycle required by the Fundamentals of Data Science (FDS) curriculum: resource verification, problem formulation, exploratory data analysis, leakage-free preprocessing, model training, multi-metric evaluation, statistical coefficient interpretation, and deployment via Streamlit.

4. Problem Statement

Let each passenger $i \in \{1, 2, \ldots, N\}$ be represented by a feature vector $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})^T$ recording pre-disaster demographic and travel characteristics, and a binary response variable $Y_i \in \{0, 1\}$, where Y_i

$= 1$ denotes that the passenger survived the disaster and $Y_i = 0$ denotes that the passenger did not survive. The statistical learning problem is to estimate the conditional probability of survival $P(Y_i = 1 \mid \mathbf{x}_i)$ using a parametric logistic regression function and assign a binary class prediction $\hat{Y}_i \in \{0, 1\}$ based on a probability decision threshold $\tau = 0.50$.

- **Dependent (Target) Variable:** `Survived` (0 = Did Not Survive, 1 = Survived).
- **Retained Independent (Predictor) Variables:** `Pclass` (ticket class), `Sex` (biological sex), Age (age in years), `SibSp` (number of siblings/spouses aboard), `Parch` (number of parents/children aboard), `Fare` (ticket fare in British pounds), and `Embarked` (port of embarkation).
- **Excluded Identifiers & Sparse Fields:** `PassengerId` (arbitrary row sequence), `Name` (unique text identifier), `Ticket` (high-cardinality alphanumeric code), and `Cabin` (77.10% missing values).

Methodological Distinction (Association vs. Causation): Throughout this study, we distinguish strictly between (1) *exploratory bivariate patterns* observed in raw group averages, (2) *multivariable statistical associations* (partial log-odds coefficients holding other covariates constant), and (3) *out-of-sample predictive accuracy* on held-out test records. Observational correlations in historical manifest data reflect historical triage rules and structural confounding rather than randomized causal interventions.

5. Objectives

1. **Cross-Format Dataset Audit:** Inspect both faculty-supplied files (`titanic.csv` and `titanic (1).xlsx`) independently to verify record counts, schemas, missing values, and data-type consistency.
2. **Exploratory Data Analysis (EDA):** Quantify univariate distributions and bivariate survival relationships across numerical and categorical features using summary tables and publication-grade visualizations.
3. **Leakage-Free Preprocessing Pipeline:** Design a reproducible `scikit-learn` `ColumnTransformer` that fits imputation medians/modes, feature scalers, and one-hot encoders strictly on training data.
4. **Model Training & Validation:** Train an L2-regularized Logistic Regression classifier on a stratified 80% training split (n = 712) and evaluate stability via 5-fold stratified cross-validation.
5. **Multi-Metric Test Set Evaluation:** Assess generalization on the held-out 20% test set (n = 179) using Accuracy, Precision, Recall, Specificity, F1-score, Confusion Matrix, and the ROC-AUC curve.
6. **Coefficient & Odds-Ratio Interpretation:** Compute standardized coefficients (β_j), odds ratios (e^{β_j}), 95% confidence intervals, Wald z-statistics, and p-values, accounting for categorical reference levels.
7. **Interactive Streamlit Deployment:** Build and validate a modular Streamlit application (`app.py`) that loads the serialized pipeline (`titanic_logistic_model.joblib`) and outputs survival predictions and probabilities.

6. Dataset Description and Cross-Format Audit

The faculty provided two dataset files: `titanic.csv` and `titanic (1).xlsx`. Both files contain **891 passenger records** and **12 columns**. An automated cell-by-cell audit revealed that while both files contain identical passenger observations and identical missing-value masks, calling `df_csv.equals(df_xlsx)` directly in `pandas` returns `False`. Inspection showed that Excel serializes the 661 purely numeric `Ticket` numbers (such as `113803`) as Python `int` objects while keeping the 230

alphanumeric tickets (such as `A/5 21171`) as `str` objects. Once `Ticket` is cast to string (`df_xlsx['Ticket'].astype(str)`), the two datasets match 100% across all 891 × 12 cells. To avoid mixed-type serialization issues, `titanic.csv` is used as the primary input file.

Column Name	Data Type	Present	Missing (%)	Unique	Description & Modeling Role
`PassengerId`	`int64`	891	0 (0.00%)	891	Sequential row identifier (1 to 891). **Excluded** (non-predictive index).
`Survived`	`int64`	891	0 (0.00%)	2	Binary target outcome (0 = Did Not Survive [549], 1 = Survived [342]). **Target (y)**.
`Pclass`	`int64`	891	0 (0.00%)	3	Ticket class / socioeconomic tier (1 = 1st [216], 2 = 2nd [184], 3 = 3rd [491]). **Retained**.
`Name`	`object`	891	0 (0.00%)	891	Passenger full name and title. **Excluded** (unique text string per row).
`Sex`	`object`	891	0 (0.00%)	2	Biological sex (`male` = 577 [64.76%], `female` = 314 [35.24%]). **Retained**.
`Age`	`float64`	714	177 (19.87%)	88	Age in years (0.42 to 80.0; mean = 29.70, median = 28.00). **Retained (Median Imputed)**.
`SibSp`	`int64`	891	0 (0.00%)	7	Number of siblings or spouses aboard (0 to 8; mean = 0.52). **Retained**.
`Parch`	`int64`	891	0 (0.00%)	7	Number of parents or children aboard (0 to 6; mean = 0.38). **Retained**.
`Ticket`	`object`	891	0 (0.00%)	681	Ticket code string. **Excluded** (high cardinality: 681 unique codes).
`Fare`	`float64`	891	0 (0.00%)	248	Passenger ticket fare in £ (0.00 to 512.33; mean = 32.20, median = 14.45). **Retained**.
`Cabin`	`object`	204	687 (77.10%)	147	Cabin number string. **Excluded** (77.10% missingness; severe sparsity).
`Embarked`	`object`	889	2 (0.22%)	3	Boarding port (`S` = 644, `C` = 168, `Q` = 77). **Retained (Mode Imputed)**.

7. Methodology

The methodology follows a strict six-stage supervised learning protocol designed to guarantee reproducibility and eliminate data leakage:

1. **Data Integrity & Quality Audit:** Check exact duplicate rows (0 found), duplicate `PassengerId` or `Name` entries (0 found), missing-value distributions, and zero-fare records (15 complimentary/work tickets found and retained as valid historical observations).
2. **Exploratory Data Analysis:** Compute parametric (mean, standard deviation) and non-parametric (median, interquartile range) statistics alongside Pearson correlations and group-level survival rates.

3. **Target Separation & Stratified Splitting:** Separate `Survived` (y) from the 7 retained predictors (X) prior to any transformation, then partition the dataset into an 80% training set ($n = 712$) and a 20% test set ($n = 179$) using `train_test_split(..., stratify=y, random_state=42)`.
4. **Leakage-Free Transformation via `ColumnTransformer`:** Encapsulate numerical median imputation and `StandardScaler` normalization alongside categorical mode imputation and `OneHotEncoder(drop='first')` inside a unified `scikit-learn` `Pipeline`. All transformer parameters are estimated exclusively from the 712 training observations.
5. **Model Fitting & Cross-Validation:** Fit an L2-penalized Logistic Regression classifier (`C=1.0`, `solver='lbfgs'`) on the training set and validate stability across 5 stratified folds.
6. **Out-of-Sample Evaluation, Statistical Inference & Deployment:** Compute classification metrics, ROC-AUC, and Wald z-tests on the test set and serialize the pipeline via `joblib` for the Streamlit application.

8. Exploratory Data Analysis (EDA) and Statistical Findings

Descriptive statistics for all numerical features across the 891 passengers are presented below, followed by group-level survival breakdowns across categorical dimensions.

Numerical Feature	Count	Mean	Std. Dev.	Min	25% (Q1)	50% (Median)	75% (Q3)	Max
`Survived` (Target)	891	0.3838	0.4866	0.0000	0.0000	0.0000	1.0000	1.0000
`Pclass` (Ticket Class)	891	2.3086	0.8361	1.0000	2.0000	3.0000	3.0000	3.0000
Age (Years)	714	29.6991	14.5265	0.4200	20.1250	28.0000	38.0000	80.0000
`SibSp` (Siblings/Spouses)	891	0.5230	1.1027	0.0000	0.0000	0.0000	1.0000	8.0000
`Parch` (Parents/Children)	891	0.3816	0.8061	0.0000	0.0000	0.0000	0.0000	6.0000
`Fare` (British Pounds £)	891	32.2042	49.6934	0.0000	7.9104	14.4542	31.0000	512.3292

Key Exploratory Findings:

- **Target Class Balance:** Exactly **549 passengers (61.62%) perished** and **342 passengers (38.38%) survived**, establishing a baseline majority-class accuracy of 61.62%.
- **Biological Sex Disparity:** Female passengers achieved a **74.20% survival rate** (233/314), compared to **18.89%** (109/577) for male passengers—a 3.93-fold relative survival ratio.
- **Socioeconomic Ticket Class Gradient:** Survival declined monotonically with ticket class: **62.96% in 1st Class** (136/216), **47.28% in 2nd Class** (87/184), and **24.24% in 3rd Class** (119/491). Within sex-by-class subgroups, 96.81% of 1st-class females and 92.11% of 2nd-class females survived, whereas only 50.00% of 3rd-class females and 13.54% of 3rd-class males survived.
- **Age & Child Priority:** Although overall median age was 28.0 years for both survivors (mean = 28.34) and non-survivors (mean = 30.63), children aged ≤12 years had a **57.97% survival rate** (40/69), whereas seniors aged >60 had only a **22.73% survival rate** (5/22).

- **Fare & Embarkation Confounding:** Survivors paid more than double the average ticket fare of non-survivors (£48.40 vs. £22.12). Passengers boarding at Cherbourg (C) had a 55.36% survival rate (93/168) compared to 38.96% at Queenstown (Q, 30/77) and 33.70% at Southampton (S, 217/644), largely because half of Cherbourg passengers traveled in 1st class.

9. Data Preprocessing Pipeline

To prevent data leakage—a common methodological flaw in which test-set statistics contaminate model training—all preprocessing transformations were implemented inside a `scikit-learn` `ColumnTransformer` fit solely on the 712 training rows:

- **Missing Numerical Imputation (Age):** The 177 missing Age records (19.87%) are imputed using `SimpleImputer(strategy='median')`. Because Age and `Fare` exhibit positive skewness, the training median (**28.50 years** for Age, **£14.4542** for `Fare`) provides a more robust central estimate than the arithmetic mean.
- **Missing Categorical Imputation (`Embarked`):** The 2 missing `Embarked` records (0.22%) are imputed using `SimpleImputer(strategy='most_frequent')`, which assigns the training mode (Southampton, `'S'`).
- **Categorical Encoding:** Nominal features `Sex` and `Embarked` are transformed via `OneHotEncoder(drop='first', sparse_output=False)`. Dropping the first lexicographical level establishes `Sex = female` and `Embarked = C` (Cherbourg) as reference categories and yields three binary indicator columns: `Sex_male`, `Embarked_Q`, and `Embarked_S`.
- **Feature Standardization:** All five numeric predictors (`Pclass`, `Age`, `SibSp`, `Parch`, `Fare`) are standardized using `StandardScaler()` ($z_j = (x_j - \mu_{train,j}) / \sigma_{train,j}$). Standardization ensures that L2 regularization penalizes all continuous features equitably regardless of measurement units (years vs. British pounds) and allows direct comparison of standardized coefficient magnitudes.
- **Stratified Splitting Verification:** The 80/20 stratified split preserves the exact target survival proportion across subsets: **38.34%** survived in the training set (273/712) and **38.55%** survived in the test set (69/179).

10. Logistic Regression Model

Logistic Regression models the log-odds (logit) of passenger survival as a linear function of the 8 preprocessed features $\mathbf{z}_i = (z_{i1}, \ldots, z_{i8})^T$:

$$ln[\, P(Y_i = 1 \mid \mathbf{z}_i) / (1 - P(Y_i = 1 \mid \mathbf{z}_i)) \,] = \beta_0 + \beta_1 z_{i1} + \beta_2 z_{i2} + \ldots + \beta_8 z_{i8}$$

Applying the inverse logit—the **sigmoid function** $\sigma(\eta) = 1 / (1 + e^{-\eta})$—maps the unbounded linear predictor $\eta_i = \beta_0 + \mathbf{z}_i^T \boldsymbol{\beta} \in (-\infty, +\infty)$ smoothly onto the unit probability interval $(0, 1)$. When the linear log-odds score $\eta_i = 0$, the predicted survival probability is exactly $\sigma(0) = 0.50$. Model parameters are estimated by minimizing the L2-regularized binary cross-entropy loss function ($C = 1.0$) via the L-BFGS-B optimization algorithm:

$$L(\beta_0, \boldsymbol{\beta}) = -\sum_{i=1}^{N_{train}} [\, y_i \ln(\hat{p}_i) + (1 - y_i) \ln(1 - \hat{p}_i) \,] + (1 / 2C) \sum_{j=1}^{8} \beta_j^2$$

Across 5-fold stratified cross-validation on the training set ($n = 712$, evaluated via `cross_validate` with `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)` in Python 3.11.2 / `scikit-learn 1.5.2`), the pipeline achieved a mean CV Accuracy of **79.49% (±2.73%)** (fold scores:

[0.7972, 0.7972, 0.8028, 0.7465, 0.8310]), mean CV ROC-AUC of **0.8549 (±0.0212)** (fold scores: [0.8569, 0.8268, 0.8762, 0.8351, 0.8796]), and mean CV F1-score of **0.7246 (±0.0335)** (fold scores: [0.7339, 0.7071, 0.7358, 0.6727, 0.7736]), demonstrating stable generalization without overfitting.

11. Evaluation Metrics and Results

The trained pipeline was evaluated on the held-out test set ($n = 179$). Every metric is reported below alongside its mathematical definition, Titanic domain interpretation, and diagnostic limitation.

Evaluation Metric	Formula	Train (n=712)	Test (n=179)	Domain Interpretation & Limitation
Accuracy	$(TP + TN) / N$	80.76%	**80.45%**	144 of 179 test passengers classified correctly (+18.83 percentage points above the 61.62% majority baseline). *Limitation:* Masks class-specific error asymmetry under class imbalance.
Precision (Class 1)	$TP / (TP + FP)$	76.98%	**79.31%**	When the model predicts a passenger survived, it is correct 79.31% of the time (46/58). *Limitation:* Ignores missed survivors (false negatives).
Recall / Sensitivity	$TP / (TP + FN)$	71.06%	**66.67%**	The model correctly identifies 66.67% (46/69) of actual test-set survivors. *Limitation:* 23 survivors (mostly 2nd/3rd class men) are missed at threshold 0.50.
Specificity (TNR)	$TN / (TN + FP)$	86.79%	**89.09%**	The model correctly identifies 89.09% (98/110) of non-survivors. Shows stronger performance on the majority non-surviving class.
F1-Score (Class 1)	$2PR / (P + R)$	0.7390	**0.7244**	Harmonic mean balancing Precision (79.31%) and Recall (66.67%) for the minority surviving class.
ROC-AUC Score	Area under TPR vs. FPR curve	0.8594	**0.8433**	Threshold-independent ranking metric: 84.33% probability that a randomly chosen survivor receives a higher score than a non-survivor.

Confusion Matrix & Error Analysis (Train $n = 712$ & Test $n = 179$): On the training set ($n = 712$), the classifier yields **381 True Negatives**, **58 False Positives**, **79 False Negatives**, and **194 True Positives** (Precision = 76.98%, Recall = 71.06%, Specificity = 86.79%, F1 = 0.7390). On the held-out test set ($n = 179$) at the default threshold ($\tau = 0.50$), the model yields **98 True Negatives (TN, 54.75%)**, **46 True Positives (TP, 25.70%)**, **12 False Positives (FP, 6.70%)**, and **23 False Negatives (FN, 12.85%)**. False Positives represent passengers whose demographic profile predicted survival (such as 1st/2nd class women or children in large families) but who perished during the sinking. False Negatives represent actual survivors whom the model predicted would perish—predominantly adult male passengers in 2nd and 3rd class who beat the statistical odds by boarding collapsible lifeboats or being pulled from the water.

12. Graphs and Visual Interpretations

All eight figures generated during the study are presented below with complete analytical interpretations.

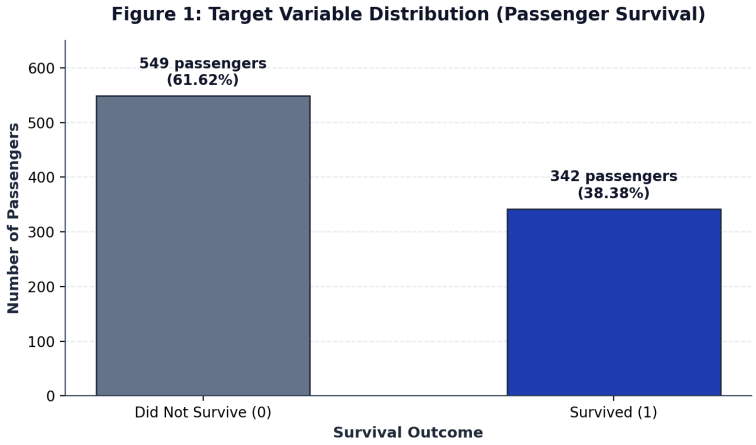


Figure 1: Target Variable Distribution (Survived). The dataset exhibits a 61.62% (549) to 38.38% (342) split favoring non-survival. Because the classes are moderately imbalanced, evaluating accuracy alone is insufficient; stratified train-test splitting and F1/ROC-AUC metrics are required.

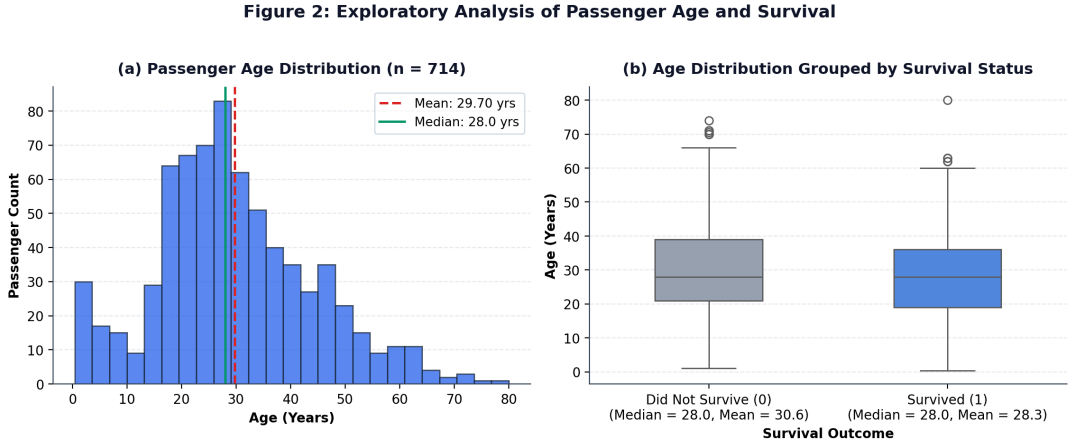


Figure 2: Passenger Age Histogram (a) and Box Plot by Survival Status (b). Observed passenger ages (n = 714) follow a unimodal, slightly right-skewed distribution with mean 29.70 years and median 28.00 years. Panel (b) shows that while median age is 28.0 years in both groups, the surviving distribution extends lower into infancy/early childhood, reflecting child prioritization during lifeboat loading.

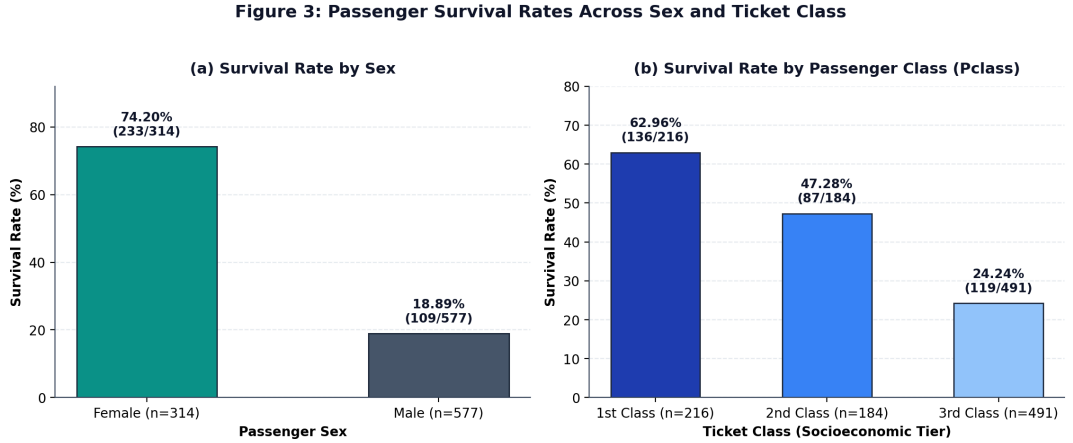


Figure 3: Bivariate Survival Rates by Sex (a) and Ticket Class (b). Panel (a) highlights the dominant predictor in the dataset: 74.20% of females survived compared to 18.89% of males. Panel (b) demonstrates a strong socioeconomic gradient: 62.96% survival in 1st class vs. 47.28% in 2nd class and 24.24% in 3rd class.

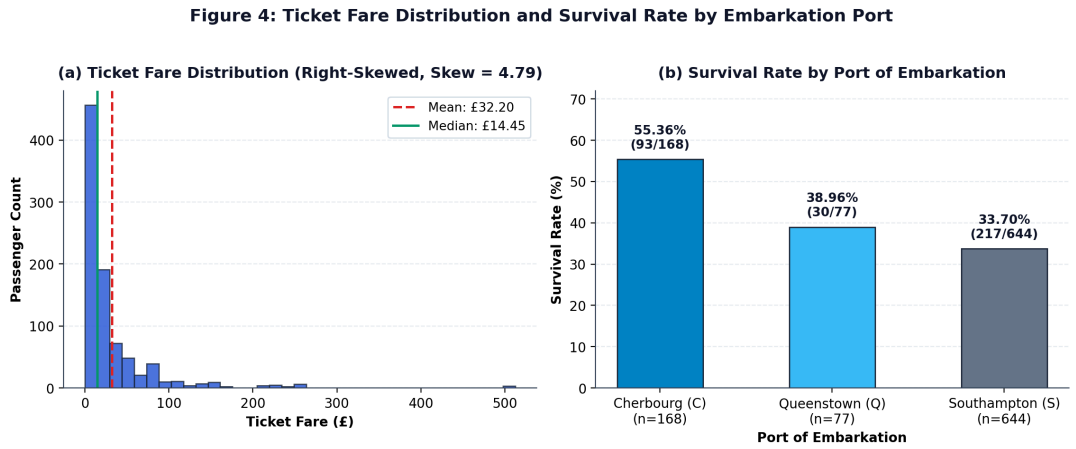


Figure 4: Ticket Fare Distribution (a) and Survival Rate by Port of Embarkation (b). Ticket fares are heavily right-skewed (mean £32.20 vs. median £14.45, max £512.33). Passengers boarding at Cherbourg (C) experienced a 55.36% survival rate compared to 38.96% at Queenstown (Q) and 33.70% at Southampton (S), driven by the concentration of 1st-class passengers embarking at Cherbourg.

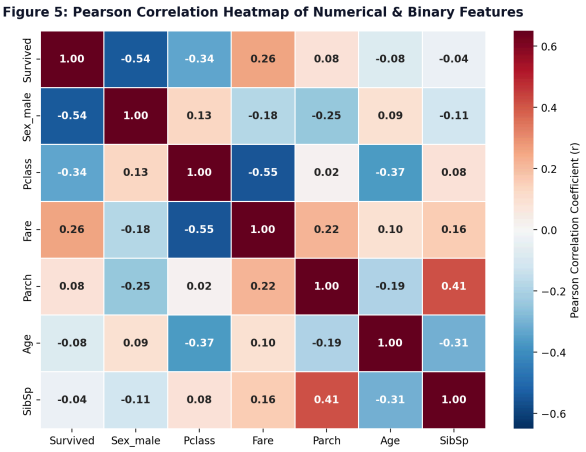


Figure 5: Pearson Correlation Heatmap of Numerical and Binary Features. Survival correlates most strongly with `Sex_male` (r = −0.54), `Pclass` (r = −0.34), and `Fare` (r = +0.26). Moderate collinearity exists between `Pclass` and `Fare` (r = −0.55) and between family size variables `SibSp` and `Parch` (r = +0.41).

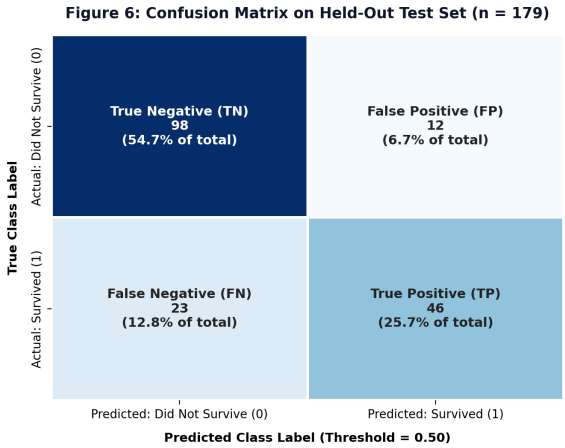


Figure 6: Annotated Confusion Matrix on the Held-Out Test Set (n = 179). The model correctly classifies 98 of 110 non-survivors (89.09% specificity) and 46 of 69 survivors (66.67% sensitivity), with 12 False Positives and 23 False Negatives.

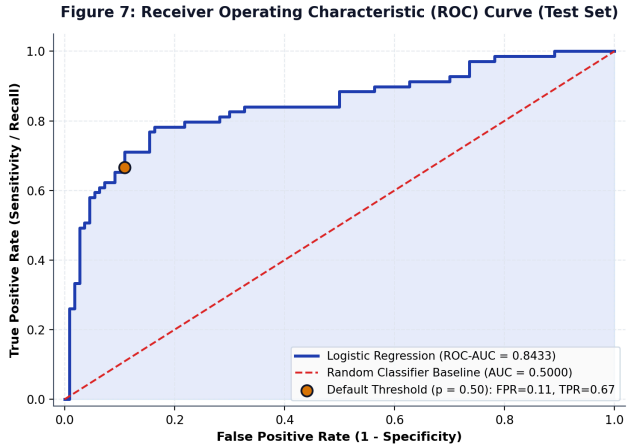


Figure 7: Receiver Operating Characteristic (ROC) Curve on the Held-Out Test Set. The ROC curve bows steeply toward the upper-left corner, achieving an area under the curve (ROC-AUC) of 0.8433 compared to the 0.5000 diagonal random baseline. The orange marker denotes the default p = 0.50 operating threshold (FPR = 0.11, TPR = 0.67).

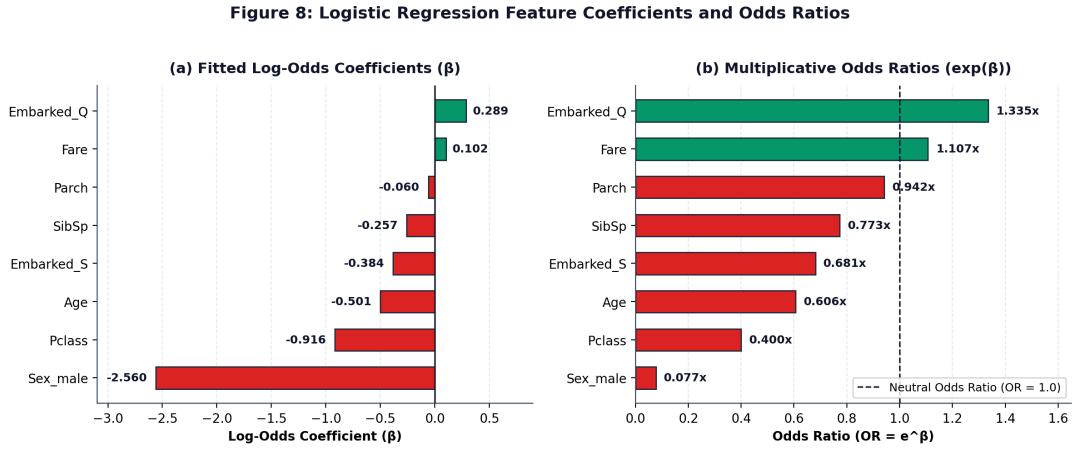


Figure 8: Fitted Logistic Regression Coefficients (β, Panel a) and Multiplicative Odds Ratios (e^{β}, Panel b). Negative log-odds coefficients (red bars, OR < 1.0) reduce survival odds, led by `Sex_male` (β = −2.560, OR = 0.077x), `Pclass` (β = −0.916, OR = 0.400x), Age (β = −0.501, OR = 0.606x), and `SibSp` (β = −0.257, OR = 0.773x).

13. Coefficient and Odds-Ratio Interpretation

Table 5 reports the fitted intercept (β_0 = *+1.2568, SE = 0.2524*) and all eight feature coefficients after preprocessing. Because numerical features were standardized via `StandardScaler`, each numerical β_j measures the change in log-odds per **one-standard-deviation increase** in that predictor, whereas categorical β_j values measure the log-odds shift of the indicator category relative to its omitted reference level (`Sex = female`, `Embarked = C`).

Transformed Feature	Reference / Unit Scale	Coef (β)	Std. Error	Wald z	p-value	Odds Ratio (e^{β})	95% CI for OR
`Sex_male`	Ref: `female` (0 → 1)	−2.5600	0.2132	−12.01	<0.0001	**0.0773**	[0.0509, 0.1174]
`Pclass`	1 SD (+0.833 class tier)	−0.9158	0.1321	−6.93	<0.0001	**0.4002**	[0.3089, 0.5185]
Age	1 SD (+13.02 years)	−0.5007	0.1101	−4.55	<0.0001	**0.6061**	[0.4884, 0.7522]

Transformed Feature	Reference / Unit Scale	Coef (β)	Std. Error	Wald z	p-value	Odds Ratio (e^{β})	95% CI for OR
Embarked_S	Ref: Cherbourg (C)	−0.3841	0.2466	−1.56	0.1192	0.6810	[0.4200, 1.1042]
Embarked_Q	Ref: Cherbourg (C)	+0.2891	0.3957	+0.73	0.4650	1.3353	[0.6148, 2.9000]
SibSp	1 SD (+1.060 siblings/spouses)	−0.2573	0.1250	−2.06	0.0396	**0.7731**	[0.6051, 0.9878]
Fare	1 SD (+£48.03 fare)	+0.1019	0.1358	+0.75	0.4532	1.1072	[0.8485, 1.4448]
Parch	1 SD (+0.838 parents/children)	−0.0596	0.1035	−0.58	0.5643	0.9421	[0.7692, 1.1539]

Distinguishing Coefficient Magnitude from Statistical Significance: A critical analytical insight emerges when comparing Embarked_S ($\beta = -0.3841$) and Embarked_Q ($\beta = +0.2891$) against SibSp ($\beta = -0.2573$). Although the point coefficients on Embarked_S and Embarked_Q appear larger in absolute magnitude than SibSp, binary indicator variables have larger standard errors and different unit scales than standardized continuous variables. Formal Wald tests show that SibSp is statistically significant at $\alpha = 0.05$ ($z = -2.06$, $p = 0.0396$), whereas Embarked_S ($p = 0.1192$), Embarked_Q ($p = 0.4650$), Fare ($p = 0.4532$), and Parch ($p = 0.5643$) have 95% confidence intervals that cross the neutral odds ratio of 1.0. Once Pclass is controlled for, ticket fare and embarkation port add little independent linear signal.

14. Discussion, Limitations, and Streamlit Deployment

Practical Relevance & Streamlit Application (app.py): To translate the statistical model into an accessible decision-support artifact, we developed a modular Streamlit web application (app.py). The application loads the serialized pipeline (models/titanic_logistic_model.joblib), validates user-supplied passenger inputs (Pclass, Sex, Age, SibSp, Parch, Fare, Embarked), computes both the predicted class and exact survival probability via predict_proba(), and displays an interactive probability progress bar alongside model diagnostics. For example, a 28-year-old 1st-class female boarding at Cherbourg (Fare = £80) receives a **95.28% survival probability** (Survived = 1), whereas a 22-year-old 3rd-class male boarding at Southampton (SibSp = 1, Fare = £7.25) receives a **9.08% survival probability** (Survived = 0).

Key Methodological Limitations:

- **Additive Linearity in Log-Odds:** Standard logistic regression assumes effects are additive on the logit scale. In reality, historical survival exhibited an interaction between Sex and Pclass: 1st- and 2nd-class women had >92% survival, whereas 3rd-class women had 50.0% survival.
- **Missing Age Imputation Compression:** Imputing 177 missing ages (19.87%) with the training median (28.5 years) preserves sample size for training but attenuates age variance.
- **Unobserved Historical Confounders:** Exact cabin corridor location at the time of impact, swimming ability, and whether a passenger approached the port vs. starboard lifeboat deck (where officers enforced different loading rules) are absent from the dataset.

15. Conclusion

This study developed, verified, and deployed an end-to-end Logistic Regression pipeline for predicting passenger survival on the RMS *Titanic*. By combining rigorous cross-format dataset auditing, leakage-free `ColumnTransformer` preprocessing, stratified splitting, and multi-metric evaluation, the model achieved **80.45% test accuracy** and an **ROC-AUC of 0.8433**. Coefficient and odds-ratio analyses confirmed that sex (`Sex_male` OR = 0.0773), ticket class (`Pclass` OR = 0.4002), age (Age OR = 0.6061), and sibling/spouse count (`SibSp` OR = 0.7731) are the primary statistically significant predictors of survival.

16. References

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